

Mycelial Memories: A Living Archive of Human Civilization

Azure Qianwen Zhang*

Hong Kong University of Science and Technology (Guangzhou), China



Figure 1: Screenshot from VR Recording of *Mycelial Memories* © Azure Qianwen Zhang 2025

ABSTRACT

Mycelial Memories is an immersive virtual reality installation that reimagines mycelial networks as living archives bridging temporal boundaries through immersive VR experiences. Drawing on anthropomorphism as a methodological framework, this work translates non-human fungal perception into human-perceivable language, creating an experiential meditation on memory, time, and ecological interconnection. Participants navigate a bioluminescent digital forest where diverse mushroom species preserve and present fragments of human civilization across AI-mediated past, present and future. By positioning mycelial networks as alternative models of documentation and memory, the installation challenges anthropocentric narratives and proposes a multispecies perspective to understanding human history within broader ecological timescales.

Index Terms: Virtual Reality, Anthropomorphism, Mycelial Networks, AI Art, Post-Anthropocentrism, Temporal Experience, Digital Memory.

1 INTRODUCTION

How might non-human living systems perceive and archive human civilization? This question forms the conceptual foundation of *Mycelial Memories*, a virtual reality installation that em-

ployes anthropomorphism as an artistic strategy to examine alternative modes of historical documentation and temporal experience. By reimagining fungi as conscious archivists, the work bridges human and non-human perspectives, creating what Haraway describes as "sympoietic" relationships—systems that make each other through collaborative becoming[7]. The installation draws inspiration from mycelial networks, the vast underground fungal systems that connect forest ecosystems through biochemical and electrical communication[9]. These networks function as distributed intelligence systems, sharing nutrients and information across species boundaries. By translating this biological model into immersive digital space, *Mycelial Memories* proposes the organic system of mycelial network as metaphorical interfaces between natural wisdom and civilization footprint, between decay and renewal, between past and future. This paper examines the theoretical framework underlying the installation, focusing on anthropomorphism as a critical methodology for translating non-human perception and the role of AI empowered XR in creating empathetic bridges across species and time.

2 THEORETICAL FRAMEWORK

2.1 Anthropomorphism as Artistic Methodology

Anthropomorphism—the attribution of human characteristics to non-human entities—has historically been dismissed as a cognitive error or sentimental projection. However, recent scholarship reconsiders anthropomorphism as a productive creative strategy. Guthrie argues that anthropomorphism functions as a perceptual tool for making sense of ambiguous phenomena[6]. Anthropomorphic strategies in robotic and technological art create bridges

*e-mail: qzhang861@connect.hkust-gz.edu.cn

between human understanding and machine logic, enabling audiences to engage with non-human systems through familiar cognitive frameworks[5]. Mycelial Memories extends this approach to natural systems, using anthropomorphic framing not to reduce fungi to human terms, but to create productive tension between human and fungal modes of being. This aligns with new materialist perspectives that recognize the agency of non-human matter while acknowledging the inevitable mediation of human perception. As Bennett argues, all matter possesses “thing-power”—vitality and agency independent of human attribution[1]. The challenge for artistic practice becomes how to acknowledge this agency while making it accessible to human audiences. Anthropomorphism, when employed critically and reflexively, offers one solution to this paradox.

2.2 Post-Anthropocentric Perspectives

The choice of fungi as conceptual anchors is deliberate. As Sheldrake documents, fungi exist at the intersection of life and death, breaking down organic matter while enabling new growth[9]. They are the natural archivists, decomposing the past to nourish the future. This positions them as ideal metaphorical interfaces for examining temporal continuity and transformation. The installation positions itself within post-anthropocentric thought, which challenges human exceptionalism and recognizes humans as embedded within rather than separate from ecological systems. Braidotti’s concept of the posthuman subject emphasizes relationality and becoming across species boundaries[2]. By presenting human history through fungal consciousness, Mycelial Memories challenges anthropocentric narratives, encouraging what Morton calls “ecological awareness”—a recognition of interconnection and shared fate[8].

3 ARTWORK DESCRIPTION

3.1 AI as Anthropomorphism Agent

Artificial intelligence systems can be understood as mirrors of human society: the responses they generate, the associations they form, and the knowledge they prioritize are shaped by patterns embedded in their training data, reflecting existing cultural values, biases, and power structures rather than autonomous intelligence[3]. The AI component in this work translates between non-human pattern recognition (the computational “perception” of historical trends) and human-perceivable imagery. This creates a layered anthropomorphic structure—fungi anthropomorphized as archivists, AI anthropomorphized as pattern-finder, both mediating between non-human processes and human understanding.

3.2 Visual and Audio Aesthetics

The virtual environment manifests as a bioluminescent forest ecosystem populated by diverse fungal species. Each species evolves specialized characteristics reflecting its archival function: industrial-scarred fungi preserve memories of urbanization and technological development; delicate crystalline varieties hold cultural celebrations and artistic expressions; decomposer species reveal environmental degradation and ecological renewal cycles. As participants navigate the forest, approaching mushrooms triggers the emergence of documentary fragments from critical moments in human civilization as projections in front of the participants. These range from archival footage of vanished cultural practices and ecological knowledge to contemporary urban development and climate events. The interaction model emphasizes embodied exploration rather than goal-directed navigation. Participants create personal pathways through collective history, making choices about which memories to investigate and connect. Immersive spatial audio reinforces the networked intelligence concept. Sounds propagate through the mycelial network, creating acoustic ecology where historical voices, contemporary urban soundscapes, and synthesized future tones blend and interfere. This sonic layering emphasizes

temporal coexistence rather than succession, with multiple time periods audible simultaneously depending on the participant’s position within the network.

4 XR AND NON-LINEAR TEMPORAL REALITY

The work positions XR technology not as escape from nature but as a tool for deepening ecological connection. Rather than treating virtual reality as dematerialized fantasy, the installation uses digital space to manifest biological principles and material relationships. By collapsing linear time into spatial, networked topology, this VR installation challenges conventional temporal experience. Past, present, and future coexist and influence each other simultaneously, reflecting ecological and systems-thinking approaches to time. This resonates with Ernst’s concept of “temporal media,” where digital systems enable non-linear temporal structures[4]. Experiencing time as mycelial network rather than human chronology encourages consideration of longer timescales and broader contexts. Ecological processes operate across centuries; civilizational impacts extend far beyond individual lifetimes.

5 CONCLUSION

Mycelial Memories proposes anthropomorphism as rigorous artistic methodology when employed reflexively and critically. The mycelial network serves as both metaphor and method—a way of organizing memory that emphasizes connection over hierarchy. By translating AI mediated fungal intelligence into immersive VR experiences, the installation creates productive spaces for examining human civilization from non-human perspectives, challenging anthropocentric views and provoking the imagination of multispecies entanglement. It demonstrates how immersive technologies might serve not just as entertainment or information delivery, but as tools for cultivating ecological awareness and multispecies empathy.

REFERENCES

- [1] J. Bennett. *Vibrant Matter: A Political Ecology of Things*. Duke University Press, Durham, NC, 2010. 2
- [2] R. Braidotti. *The Posthuman*. Polity Press, Cambridge, 2013. 2
- [3] K. Crawford. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press, New Haven, CT, 2021. 2
- [4] W. Ernst. *Digital Memory and the Archive*. University of Minnesota Press, Minneapolis, 2013. 2
- [5] F. Ghedini and M. Bergamasco. Robotic creatures: Anthropomorphism and interaction in contemporary art. In *Proceedings of the IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 2010. doi: 10.1109/ROMAN.2010.5598720 2
- [6] S. E. Guthrie. *Faces in the Clouds: A New Theory of Religion*. Oxford University Press, Oxford, 1993. 1
- [7] D. J. Haraway. *Staying with the Trouble: Making Kin in the Chthulucene*. Duke University Press, Durham, NC, 2016. 1
- [8] T. Morton. *Dark Ecology: For a Logic of Future Coexistence*. Columbia University Press, New York, 2016. 2
- [9] M. Sheldrake. *Entangled Life: How Fungi Make Our Worlds, Change Our Minds, and Shape Our Futures*. Random House, New York, 2020. 1, 2